Effectiveness of Treatments for Metastatic Uveal Melanoma

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• PURPOSE: To evaluate and comment on published peer-reviewed literature for evidence of effectiveness of treatments for metastatic uveal melanoma.

• DESIGN: Analytical nonexperimental study of published peer-reviewed data.

• METHODS: Literature search and analysis of pertinent articles published between January 1, 1980 and June 30, 2008.

• RESULTS: Of 80 identified publications, 12 (15.0%) were review articles without original information, 2 (2.5%) were review articles combined with case reports, 22 (27.5%) were case reports, 16 (20.0%) were retrospective descriptive case series reports, 3 (3.75%) were pilot studies of a novel intervention, 2 (2.5%) were prospective phase I clinical trials, 8 (10.0%) were prospective phase I/II clinical trials, and 15 (18.75%) were prospective phase II clinical trials. None of the articles reported a prospective, randomized phase III clinical trial. The largest reported unselected patient groups had a median survival of 3 to 4 months after detection of metastasis, whereas the largest selected patient groups showed substantially longer median survival times.

• CONCLUSIONS: Although median survival time after diagnosis of metastatic uveal melanoma tends to be substantially longer in selected patient subgroups subjected to aggressive invasive interventions than it is in unselected groups, much if not most of this apparent difference in survival is likely to be attributable to selection bias, surveillance bias, and publication bias rather than treatment-induced alteration of expected outcome. Published peer-reviewed articles do not provide compelling scientific evidence of any survival benefit of any method of treatment for any subgroup of patients with metastatic uveal melanoma. (Am J Ophthalmol 2009;148:119–127. © 2009 by Elsevier Inc. All rights reserved.)

PATIENTS IN WHOM METASTASIS DEVELOPS AS A result of primary uveal melanoma rarely survive more than a few years after initial detection of that metastasis. The median survival time after diagnosis of metastasis in the largest published series of unselected patients with metastatic uveal melanoma was 3.6 months.[1] Cumulative actuarial survival in this series was 20% at 1 year, 13% at 2 years, 5% at 3 years, 2% at 4 years, and less than 1% at 5 years. Despite the generally acknowledged poor prognosis of patients with metastatic uveal melanoma, one occasionally encounters an article that purports to show substantially prolonged survival of selected patients who were subjected to certain aggressive interventions, most notably surgical resection of metastatic tumors (metastectomy) and various techniques of regional perfusion chemotherapy.[2–5] Despite such claims, many oncology groups and individual clinicians remain unconvinced about the beneficial impacts of currently available treatments on survival time.[6–8]

The current study was designed to evaluate the quality of evidence[9] regarding the impact of treatment on overall survival time in patients with metastatic uveal melanoma that has been published in the peer-reviewed literature during the past quarter century. It is a slight expansion and modification of a paper presented by the authors at the 2008 American Ophthalmological Society Meeting.[10] Four specific aspects of quality (see METHODS SECTION for details) were addressed in this study: 1) type of report, 2) category of reported subgroup, 3) baseline clinical variables evaluated and reported, and 4) control or comparison group reported.

METHODS

THE AUTHORS PERFORMED A LITERATURE SEARCH TO identify all original articles on metastatic uveal melanoma that were published in peer-reviewed medical journals during the period January 1, 1980 through June 30, 2008. The first step in this search was to perform a PubMed search evaluating the terms *uveal melanoma*, *choroidal melanoma*, *ciliary body melanoma*, *ciliochoroidal melanoma*, *iris melanoma*, *iridociliary melanoma*, *intraocular melanoma*, and *ocular melanoma*. Articles dealing with laboratory investigations or using animal models were excluded. The set of articles identified by this preliminary search then was evaluated for the additional terms *metastasis* and *metastatic disease*. The title and abstract (if provided by PubMed) of each of the articles identified at this step was reviewed by one of the authors (J.J.A.) to confirm that its subject was truly metastatic uveal melanoma and to determine whether it described treatment of any sort for the meta-

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